SANJAY BOLLINI

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EDUCATION

University of Washington Jun 2028
B.S. Computer Science | Minor: Applied Mathematics | GPA: 3.75 | Dean's List Seattle, WA

Relevant Coursework: CSE 351: Computer Systems • CSE 391: Software Tools (Linux & Git) • CSE 311: Foundations of Computing • AMATH 352: Numerical Linear Algebra • MATH 208: Matrix Algebra • MATH 126: Multivariable Calculus • CSE 332: Data Structures & Parallelism(upcoming) • CSE 333: Systems Programming(upcoming)

Certifications: Claude with the Anthropic API — Anthropic (Apr 2026) • Multi-Agent Systems — Google • AWS Cloud Essentials

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL, HTML/CSS, Bash

Stack: Next.js, Vue.js, FastAPI, Node.js, PostgreSQL, REST APIs, AWS, Git, Linux

AI-Native: Claude Code, Claude API (Haiku/Sonnet), MCP (servers/tools/resources/prompts/CLAUDE.md), LoRA fine-tuning, Ollama, multi-agent orchestration

Shipping: Vitest, Playwright E2E, Zod, TDD, CI/CD concepts, technical documentation, agile

PROJECTS

AI Market Intelligence System 2025–2026
Python, PostgreSQL, Claude Haiku/Sonnet, Ollama (local LLM), yfinance, cron | github.com/Zing302/ai-market-intel

- Built from scratch: 4-agent LLM orchestration system (Analyst, Researcher, Trader, Manager) running 3x/day on 20 symbols, ingests SEC 8-K filings, 800+ daily headlines, and real-time prices into a production PostgreSQL schema. Designed every layer independently: data ingestion, agent architecture, guard model, and reporting.
- Built a pluggable LLM provider abstraction supporting both Anthropic Claude and a local Ollama instance, switchable via a single config variable with no code changes, enabling fully offline/self-hosted operation.
- 141 unit tests, 7 cron-scheduled jobs, per-source circuit breakers, and 7-day outcome backfill scoring accuracy. Fully automated

BillPilot AI 2026
Next.js 15, TypeScript, Claude API, PostgreSQL | Submitted: OpenAI × Handshake Codex Creator Challenge

- Independently designed and shipped a full-stack AI medical billing assistant — REST API, TypeScript frontend, PostgreSQL backend, and multi-step agentic pipeline (document ingestion → LLM extraction → schema-enforced output → user review). Built AI-natively with Claude Code; authored CLAUDE.md skill files and MCP connectors throughout.
- Production-ready: Zod schema validation, Vitest unit tests, Playwright E2E, recognized as a top project in the Challenge.

Quantum Password Generator 2025
Python, FastAPI, Qiskit, IBM Qiskit Aer, HTML | github.com/Zing302/quantum-password-generator

- Quantum-mechanical true-random password generator using Hadamard gates and measurement collapse. FastAPI backend + HTML frontend: full-stack, solo-built, deployable on real IBM quantum hardware.

Mindfulness4Success 2023
Vue.js, JavaScript | mindfulness4success.com

- Built and launched a wellness web app solo at 16, presented at the Youth Entrepreneur Summit, donated proceeds to Carroll ISD CS Honor Society. Live since 2023.

EXPERIENCE

Cloud Girl Jun 2026 – Present
AI Developer Advocate Intern Remote

- Designed and shipped a full MCP demo — a live dashboard powered by 5 independent MCP servers (Calendar, Journal, Weather, Market & News, Log) with tools, resources, prompts, and an agent tool-use loop with Claude. First demo delivered for Priyanka Vergadia (Google Cloud Developer Advocate).
- Built and shipped a job-market analysis tool comparing AI Developer vs. Forward Deployment Engineer roles: a proxy-based scraper (Claude Code + DataImpulse) extracting required skills across postings, plus a resume upload feature recommending the better-fit role
- Prepared the demo and script for a 13:30 tutorial LoRA fine-tuning an open-weights LLM on a free Colab GPU using Unsloth, deployed locally via Ollama with a fully local voice interface (faster-whisper + Kokoro TTS).

Alliance Endocrinology Feb 2020 – Present
IT Manager Fort Worth, TX

- Running IT solo for a medical clinic since age 14, own every layer: network infrastructure, endpoint security (Sophos), EHR systems (EClinicalWorks), HIPAA compliance, device management, and vendor relationships. No support team, no escalation path.
- Designed and maintained clinic website (HTML/CSS); implemented SEO and Google Ads strategy that measurably improved patient acquisition.

Mouri Tech Jul 2024 – May 2025
Data Engineering Intern Irving, TX

- Built SQL analytics pipelines across full SDLC — schema design, query optimization, testing, and delivery. First professional engineering environment.

RESEARCH

UT Dallas — Software Testing & Analysis Lab Jun – Aug 2024
ML & Formal Verification Research Intern Dallas, TX

- Integrated ML into static analysis tools (JayHorn, JBMC, CBMC) to automate Java program verification; designed variance analysis pipelines and presented at a lab research showcase.

UW Machine Learning Research Club Sep 2025 – Present
Spiking Neural Networks Researcher Seattle, WA

- Collaborating on bio-inspired SNN architectures for energy-efficient, low-latency inference, applying neuromorphic computing principles to real-world ML deployment.